

Row weights on the natural endpoint scale

Research corollary; independently reviewed

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Let $p_i(q) \in [0, 1]$, $1 \leq i \leq m$, be nondecreasing in the finite ordered row index q . Give the rows positive weights w_q summing to one, and suppose that

$$\sum_q w_q \prod_{i \in S} p_i(q) = \frac{1}{|S| + 1} \quad (S \subseteq [m]).$$

Write $u_i(q) = 1 - p_i(q)$ and $s_q = \sum_i u_i(q)$.

Theorem 1. *For every fixed $L < \infty$ there is a constant C_L such that every row with $0 \leq s_q \leq L$ satisfies*

$$w_q \leq C_L \left(\frac{\sqrt{s_q}}{m^{3/2}} + \frac{1}{m^2} \right).$$

The statement allows arbitrary positive row weights; it does not assume that the rows are equally weighted.

Proof. It suffices to treat sufficiently large m , depending on L ; the remaining dimensions follow from $w_q \leq 1$ after increasing C_L . We use the reviewed sharp relative balance and distribution bound from `../sharp_endpoint_relative_balance.tex`:

$$\max_i u_i \leq \frac{C(s + m^{-1})}{m}, \quad m \sum_{s_q \leq t} w_q \leq C(t + m^{-1}) \quad (t \geq 0).$$

The second inequality is the same bound after its fixed rescaling of the deficit coordinate.

Fix one index i and put $M = m - 1$, $\ell = \lfloor M/4 \rfloor$,

$$x_i = \frac{\ell}{M}(s - u_i), \quad \bar{A}_i = \mathbb{E}_B \prod_{j \in B} p_j, \quad \mu_i(q) = (\ell + 1) \bar{A}_i(q) w_q,$$

where B is a uniform ℓ -subset of $[m] \setminus \{i\}$. This is a probability measure by the mixed-moment identities. For $s \leq L$, sharp balance makes every $u_j \leq 1/2$ for sufficiently large m . Therefore every such success product obeys

$$\prod_{j \in B} (1 - u_j) \geq \exp \left(-2 \sum_{j \in B} u_j \right) \geq e^{-2L}, \quad \mu_i(q) \geq c_L m w_q. \quad (1)$$

The variable-width mass lemma in `../independent_directions/natural_scale_endpoint_profile.tex`, together with its reviewed fixed-positive-interval companion, gives

$$\mu_i\{|x_i - a| \leq \sqrt{a/m}\} \leq C_L \sqrt{a/m} \quad (T/m \leq a \leq L), \quad (2)$$

for some fixed sufficiently large T . All constants may be enlarged to cover the overlap of the variable and fixed ranges.

For $s_q \geq T'/m$, with a sufficiently large fixed T' , sharp balance implies $u_i(q)/s_q = O(1/m)$, uniformly in this range. Consequently $x_i(q) \asymp s_q$ and $x_i(q) \geq T/m$. Apply (2) at the center $a = x_i(q)$. Its interval contains row q , so (1) yields

$$w_q \leq \frac{C_L}{m} \sqrt{x_i(q)/m} \leq C_L \frac{\sqrt{s_q}}{m^{3/2}}.$$

If $s_q \leq T'/m$, the sharp distribution bound directly gives

$$mw_q \leq m \sum_{s_r \leq s_q} w_r \leq C(s_q + m^{-1}) \leq C_{T'}/m.$$

These two cases prove the theorem, including a row at $s_q = 0$. $\square$

Sharpness for arbitrary weighted tensors. The scale is attained, up to fixed constants, within the weighted comparison-tensor class. Indeed, take a positive scalar quadrature exact through degree m , use its weights as row weights, and set all m columns equal to its node value. Then every required mixed moment is a scalar moment. The Gaussian node lemma in `../independent_directions/intermediate_endpoint_sharpness.tex` gives, for every $m^{-1} \leq \lambda \leq L$, a row with

$$s \asymp \lambda, \quad w_q \geq c_L \sqrt{\lambda} m^{-3/2}.$$

Here comparability constants are independent of m and λ ; no exact prescribed numerical deficit is asserted.

The additive floor is also attained at $s = 0$. Choose $K = \lceil (m+3)/2 \rceil$ and the positive K -node uniform Gauss–Lobatto rule, which is exact through degree $2K-3 \geq m$. An elementary exactness and positivity proof is given in `../endpoint_profile_sharpness.tex`. Its endpoint cardinal polynomial at 1 is

$$\ell_1(t) = \frac{tP'_{K-1}(2t-1)}{P'_{K-1}(1)}.$$

Indeed the interior Lobatto nodes are the zeros of the numerator's Legendre derivative. Integration by parts and Legendre orthogonality give $\int_0^1 tP'_{K-1}(2t-1) dt = 1/2$, whereas $P'_{K-1}(1) = K(K-1)/2$. Thus the endpoint weight is $1/[K(K-1)] \asymp m^{-2}$. The all-equal-column tensor at this endpoint has $s = 0$, as desired.

Scope. For a comparison tensor of actual fair dice, w_q is the probability of the distinguished face. The theorem bounds each endpoint face probability on its own deficit scale. It does not assert that every specified deficit can be realized with a matching number of faces, or provide an additional total-size construction.